RODOLPHE ABOU ASSALI

Nationality: Lebanese **Born on August 4, 2000**

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Personal Website

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Education

- **2023-Present**: Joint PhD in Differential Geometry (3rd year)
  Topic: The biharmonic Steklov problem for differential forms
  Supervisors: Nicolas Ginoux, Georges Habib, Samuel Tapie
  Lebanese University, Faculty of Science, Lebanon – University of Lorraine, Institut Élie Cartan, France.
- **2022–2023**: Master 2 in Fundamental and Applied Mathematics (Double degree)
  Lebanese University, Faculty of Science, Lebanon – University of Lorraine, Institut Élie Cartan, France.
- **2021–2022**: Master 1 in Mathematics
  Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2022**: University Diploma in Education
  Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2021**: Bachelor's Degree in Mathematics
  Lebanese University, Faculty of Science, Section II, Fanar, Lebanon.

Professional Experience

- **2025–2026**: Lecturer at IUT Nancy – Charlemagne – Computer Science Department
  Courses taught:
  - Descriptive Statistics (L1)
  - Numerical methods (L1)
  - Graph theory (L1)
  - Optimization Methods (L2, DACS).
- **2024–2025**: Lecturer at IUT Nancy – Charlemagne – Computer Science Department
  Courses taught:
  - Descriptive Statistics (L1)
  - Optimization Methods (L2, DACS)
  - Mathematical Complements and Modelling (L3, RA–IL).

- **2022–2023**: Master 2 Internship in Differential Geometry
  Topic: Spectral Theory of the Biharmonic Steklov Operator
  Supervisors: Nicolas Ginoux, Georges Habib, Samuel Tapie
  Lebanese University, Faculty of Science, Lebanon – University of Lorraine, Institut Élie Cartan, France.

- **2018–2022**: Private tutoring – Mathematics and Physics for middle and high school students.

Excellence Scholarships and Research Grants

- **2023–2026**: PhD scholarship from the SAFAR program of the French Embassy in Lebanon, for a PhD under the supervision of Nicolas Ginoux, Georges Habib, Samuel Tapie.

- **2026**: Orion Grant for Participation in International Conferences.

- **2025**: DrEAM Grant from the University of Lorraine under the Lorraine University of Excellence (LUE) initiative, for a two-month research stay at Sapienza University under the supervision of Alessandro Savo.

- **2023**: Master Grant from the Orion program at the University of Lorraine for a 4-month research stay at the IECL under the supervision of Nicolas Ginoux, Georges Habib, Samuel Tapie.

Preprints

- **2026**: *On the spectrum of the biharmonic Steklov operator with Neumann boundary conditions acting on forms of the unit Euclidean ball.* arXiv:2610.28181 [math.DG].

- **2026**: *Geometric eigenvalue estimates of Kuttler-Sigillito type on differential forms.* arXiv:2602.09876 [math.DG].

- **2025**: *Biharmonic Steklov problems with Neumann boundary conditions and spectral inequalities on differential forms.* arXiv:2507.05049 [math.DG].

Talks

Speaker

- **2026**: ESDG L'Aquila - Poster presenting arXiv:2507.05049
  Biharmonic Steklov problems and spectral inequalities on differential forms.

- **2026**: Phd away days at University of Luxembourg
  Steklov problems and spectral inequalities in planar domains.

- **2025**: Seminar of the department of Basic and Applied Sciences for Engineering at Sapienza University, Rome
  Biharmonic Steklov problems and spectral inequalities for differential forms.

- **2025**: Advanced PDE Seminar, Department of Mathematics and Statistics, University of Konstanz, Germany
  Biharmonic Steklov problems and spectral inequalities for differential forms.
- **2025**: Differential Geometry Seminar, IECL Nancy
  Biharmonic Steklov problems and spectral inequalities on differential forms.
- **2024**: IECL PhD Students Day, Nancy
  The Biharmonic Steklov Operator.
- **2024**: Doctoral Forum of the EDST, Lebanese University, Beirut
  The Biharmonic Steklov Problem on Differential Forms.

Attendee

- **2026**: European School of Differential Geometry (ESDG), L'Aquila.
- **2026**: Working group on the spectrum of hyperbolic surfaces, IECL Nancy.
- **2025**: Differential Geometry at L'Aquila (DG@AQ25).
- **2024**: European School of Differential Geometry (ESDG), Granada.
- **2024**: Working group on the spectrum of the Steklov operator, IECL Nancy.
- **2023**: Geometry Day, IECL Nancy.

Skills

Computer Skills:

- **Programming languages**: C, CSS, C++, HTML, MATLAB (Beginner), LaTeX.
- **Software**: Proteus 8 Professional, Tinkercad, Cisco Packet Tracer, Oracle Data Modeler (Intermediate).
- Adobe Photoshop, Illustrator, Dreamweaver, Visual Studio, VMware Workstation (Beginner).

Language Skills:

- Arabic (Native language)
- French (Advanced level)
- English (Advanced level)

Activities

- **2026-Present**: Elected PhD student representative on the IECL Council.
- **2026-Present**: Member at Société de Mathématiques Appliquées et Industrielles, (SMAI).
- **2026**: Co-organizer of the PhD away days in Luxembourg with the PhD students from the University of Luxembourg.

- **2025-Present**: Member at Société Mathématique de France, (SMF).
- **2024–Present**: Co-organizer of the PhD Students Seminar at IECL.
- **2023–Present**: University teaching training sessions – Université de Lorraine.
- **2025**: Co-organizer of the PhD away days in Nancy with the PhD students of the University of Luxembourg.
- **2025**: Workplace First Aid Certificate (SST) – Nancy.
- **2024–2025**: Co-facilitator of the Math Club at Lycée Jacques Callot, Vandœuvre-lès-Nancy.
- **2024–2025**: Co-facilitator for Math workshops at the Science Festival at FST and Jeanne d'Arc Elementary School.
- **2021–2023**: Student representative in M1 and M2.